Figure 9.6 The component parts of an agent profile

At a basic level, an *agent profile* is a set of prompts describing the agent. It may include other external elements related to actions/tools, knowledge, memory, reasoning, evaluation, planning, and feedback. The combination of these elements comprises an entire agent prompt profile.

Prompts are the heart of an agent's function. A prompt or set of prompts drives each of the agent components in the profile. For actions/tools, these prompts are well defined, but as we've seen, prompts for memory and knowledge can vary significantly by use case.

The definition of an AI agent profile is more than just a system prompt. Prompt flow can allow us to construct the prompts and code comprising the agent profile but also include the ability to evaluate its effectiveness. In the next section, we'll open up prompt flow and start using it.

9.3 Setting up your first prompt flow

Prompt flow is a tool developed by Microsoft within its Azure Machine Learning Studio platform. The tool was later released as an open source project on GitHub, where it has attracted more attention and use. While initially intended as an application platform, it has since shown its strength in developing and evaluating prompts/ profiles.

Because prompt flow was initially developed to run on Azure as a service, it features a robust core architecture. The tool supports multi-threaded batch processing, which makes it ideal for evaluating prompts at scale. The following section will examine the basics of starting with prompt flow.

9.3.1 Getting started

There are a few prerequisites to undertake before working through the exercises in this book. The relevant prerequisites for this section and chapter are shown in the following list; make sure to complete them before attempting the exercises: